

Sprint 1 Notes: HTML Fundamentals and Element Syntax

Sprint 1 Goal

The goal of Sprint 1 is to understand the basic way HTML is written.

You are not trying to build a full website yet. You are learning the small building blocks that every HTML page uses.

By the end of this sprint, you should understand:

- What HTML does
 - What elements are
 - What tags are
 - How opening and closing tags work
 - How nesting works
 - Why lowercase tag names are recommended
 - What void elements are
 - What attributes are
 - How the `img` element works
 - What boolean attributes are
 - The difference between `id` and `class`
 - Common beginner mistakes
 - How to debug simple broken HTML
-

1. What HTML Is

HTML stands for **HyperText Markup Language**.

In simple words:

HTML gives a web page its structure and content.

HTML tells the browser what each piece of content means.

For example, HTML can tell the browser:

- This is a main heading.
- This is a paragraph.
- This is an image.

- This is a container.
- This is a checkbox.

Example:

```
<h1>This is the main heading</h1>  
<p>This is a paragraph.</p>  

```

This code creates:

- A main heading
- A paragraph
- An image

HTML is mainly about **structure**, not design or behavior.

Think about a web page like this:

```
HTML = structure and content  
CSS = appearance and design  
JavaScript = behavior and interactivity
```

For Sprint 1, focus only on HTML structure.

2. HTML Elements

An **HTML element** is one piece of HTML.

Most elements have this pattern:

```
<tagname>content</tagname>
```

Example:

```
<p>I am learning HTML.</p>
```

This full line is one HTML element.

It has three parts:

```
Opening tag: <p>  
Content: I am learning HTML.  
Closing tag: </p>
```

The element starts with an opening tag and ends with a closing tag.

3. Tags

A **tag** is the part inside angle brackets.

Example:

```
<p>Hello</p>
```

The tags are:

```
<p>  
</p>
```

The text `Hello` is the content.

The whole thing is the element:

```
<p>Hello</p>
```

4. Opening Tags

An **opening tag** starts an element.

Example:

```
<h1>
```

This tells the browser:

A main heading starts here.

Complete example:

```
<h1>My Profile Page</h1>
```

The opening tag is:

```
<h1>
```

5. Closing Tags

A **closing tag** ends an element.

A closing tag looks like the opening tag, but it has a forward slash `/`.

Example:

```
</h1>
```

Complete example:

```
<h1>My Profile Page</h1>
```

The closing tag is:

```
</h1>
```

Correct paragraph:

```
<p>This is a paragraph.</p>
```

Incorrect paragraph:

```
<p>This is a paragraph.
```

Why is it incorrect?

Because the paragraph was opened but not closed.

Also incorrect:

```
<p>This is a paragraph.<p>
```

Why is it incorrect?

Because the second `<p>` is another opening tag, not a closing tag.

The correct closing tag must be:

```
</p>
```

6. Content Inside Elements

The content is what sits between the opening tag and the closing tag.

Example:

```
<h1>My Favorite Animal</h1>
```

Here:

```
Opening tag: <h1>  
Content: My Favorite Animal  
Closing tag: </h1>
```

The browser shows the content on the page.

7. Common Beginner Elements

Here are some basic HTML elements from Sprint 1.

h1

The **h1** element is used for the main heading of a page or section.

Example:

```
<h1>Main heading</h1>
```

h2

The **h2** element is used for a smaller section heading.

Example:

```
<h2>Smaller section heading</h2>
```

p

The **p** element is used for a paragraph.

Example:

```
<p>A paragraph of text.</p>
```

div

The **div** element is a generic container.

Example:

```
<div>A generic container.</div>
```

A **div** does not give special meaning by itself. It is mainly used to group content.

8. Nesting Elements

Nesting means putting one element inside another element.

Example:

```
<div>
  <h1>My Favorite Animal</h1>
  <p>I like cats because they are calm and independent.</p>
</div>
```

In this example:

- The `div` is the outer element.
- The `h1` is inside the `div`.
- The `p` is inside the `div`.

The `h1` and `p` are nested inside the `div`.

9. Correct Nesting

HTML should be nested like boxes inside boxes.

Correct:

```
<div>
  <p>This is inside the div.</p>
</div>
```

The structure is:

```
Open div
  Open p
  Close p
Close div
```

This is correct because the paragraph opens and closes completely inside the `div`.

10. Incorrect Nesting

Bad nesting:

```
<div>
  <p>This is broken.</div>
</p>
```

Why is this wrong?

Because the paragraph starts inside the `div`, but the `div` closes before the paragraph closes.

The structure is:

```
Open div
  Open p
Close div
Close p
```

That is not clean nesting.

Correct version:

```
<div>
  <p>This is fixed.</p>
</div>
```

11. Lowercase Tag Names

HTML tag names are mostly case-insensitive.

This may still work:

```
<H1>Hello</H1>
```

But the normal convention is to use lowercase:


```
<h1>Hello</h1>
```

Use lowercase because:

- It is cleaner.
- It is easier to read.
- It is the common style used by developers.
- It keeps your code consistent.

Recommended:

```
<p>This is good style.</p>
```

Not recommended:

```
<P>This is not the normal style.</P>
```

12. Void Elements

Some HTML elements do not wrap content.

These are called **void elements**.

A void element has only a start tag.

It does not have:

- Text content inside it
- Child elements inside it
- A closing tag

Example:

```

```

The `img` element is a void element.

You should not write:

```
Cat image</img>
```

That is wrong because `img` cannot contain content.

Another void element:

```
<input type="text">
```

The `input` element is also void.

For Sprint 1, remember:

```
img is void.  
input is void.  
Void elements do not have closing tags.  
Void elements do not contain text.  
Void elements do not contain child elements.
```

13. Attributes

Attributes give extra information to an element.

They are written inside the opening tag.

Basic pattern:

```
<tagname attribute="value">content</tagname>
```

Example:

```
<p class="intro">Welcome to my page.</p>
```

Here:

```
Element: p  
Attribute name: class
```

```
Attribute value: intro
Content: Welcome to my page.
```

Attributes must go inside the opening tag.

Correct:

```
<p class="intro">Welcome to my page.</p>
```

Incorrect:

```
<p>Welcome to my page.</p class="intro">
```

The incorrect version puts the attribute in the wrong place.

14. Attribute Name and Attribute Value

Most attributes use this format:

```
name="value"
```

Example:

```
<a href="https://www.freecodecamp.org">Visit freeCodeCamp</a>
```

Here:

```
a = anchor or link element
href = attribute name
https://www.freecodecamp.org = attribute value
Visit freeCodeCamp = clickable text
```

Another example:

```

```

Here:

```
img = image element  
src = attribute name  
cat.jpg = attribute value  
alt = attribute name  
A gray cat sitting on a sofa. = attribute value
```

15. The `img` Element

The `img` element displays an image on a web page.

Basic pattern:

```

```

Example:

```

```

The `img` element is void, so it does not have a closing tag.

Correct:

```

```

Incorrect:

```
</img>
```

Also incorrect:

```
<img>Dog picture</img>
```

16. The `src` Attribute

The `src` attribute tells the browser where the image file is.

Example:

```

```

Here:

```
src="cat.jpg"
```

means:

```
Use the image file named cat.jpg.
```

If the browser cannot find the file, the image will not display correctly.

17. The `alt` Attribute

The `alt` attribute gives text alternative for an image.

Example:

```

```

The `alt` text is:

```
A brown dog running through grass.
```

The `alt` attribute is useful because:

- It describes the image if the image does not load.
- It helps screen reader users understand the image.
- It gives meaning to the image.

Weak alt text:

```

```

Better alt text:

```

```

The better version explains what the image actually shows.

18. Boolean Attributes

Some attributes work by being present.

These are called **boolean attributes**.

A boolean attribute is either:

```
Present = on  
Absent = off
```

Example:

```
<input type="checkbox" checked>
```

This means the checkbox starts checked.

You do not need to write:

```
<input type="checkbox" checked="true">
```

The shorter version is better:

```
<input type="checkbox" checked>
```

19. Why `checked="false"` Is a Problem

This code looks like it should make the checkbox unchecked:

```
<input type="checkbox" checked="false">
```

But that is misleading.

Because `checked` is a boolean attribute.

The browser mainly cares that the attribute exists.

So this:

```
<input type="checkbox" checked="false">
```

still means:

```
checked is present  
therefore the checkbox is checked
```

If you want the checkbox checked, write:

```
<input type="checkbox" checked>
```

If you do not want the checkbox checked, remove the attribute:

```
<input type="checkbox">
```

20. Examples of Boolean Attributes

Here are boolean attributes mentioned in Sprint 1:

```
<input type="checkbox" checked>
```

The checkbox starts checked.

```
<input type="text" disabled>
```

The input is disabled.

```
<input type="text" readonly>
```

The input is read-only.

```
<input type="email" required>
```

The field is required.

For Sprint 1, you only need the basic rule:

If a boolean attribute is present, it is turned on.
If it is absent, it is turned off.

21. `id` and `class`

Both `id` and `class` are attributes used to label elements.

They are not the same.

Use them for different purposes.

22. The `id` Attribute

Use `id` for one unique element.

Example:

```
<h1 id="page-title">My Animal Page</h1>
```

This gives the `h1` a unique label:

```
page-title
```

An `id` should not be reused on the same page.

Bad:


```
<h1 id="title">Cats</h1>
<h1 id="title">Dogs</h1>
```

Why is this bad?

Because both headings use the same `id`.

Better:

```
<h1 id="cat-title">Cats</h1>
<h1 id="dog-title">Dogs</h1>
```

Now each `id` is unique.

23. Good `id` Values

Avoid spaces inside `id` values.

Bad:

```
<h1 id="main heading">My Page</h1>
```

Better:

```
<h1 id="main-heading">My Page</h1>
```

Good `id` values often use:

- Letters
- Numbers
- Dashes
- Underscores

Examples:

```
<h1 id="page-title">Page Title</h1>
<h2 id="section_1">Section 1</h2>
<p id="intro-text">Intro paragraph.</p>
```

Avoid this:

```
<h1 id="main heading">Main Heading</h1>
```

Because the `id` contains a space.

24. The `class` Attribute

Use `class` for reusable grouping.

Example:

```
<div class="card">Cat section</div>
<div class="card">Dog section</div>
```

Both `div` elements use the same class:

```
card
```

That is allowed.

A class can be reused on many elements.

25. Multiple Classes on One Element

An element can have more than one class.

Example:

```
<div class="card animal-card favorite"></div>
```

This `div` has three classes:

```
card
animal-card
favorite
```

Multiple classes are separated by spaces.

This is useful because one element can belong to several groups.

26. id vs class

Simple rule:

```
id = unique label for one element  
class = reusable label for many elements
```

Use `id` when one specific element needs a unique name.

Example:

```
<h1 id="page-title">My Page</h1>
```

Use `class` when many elements can share the same group name.

Example:

```
<div class="card">Card 1</div>  
<div class="card">Card 2</div>  
<div class="card">Card 3</div>
```

27. Common Mistakes to Avoid

Mistake 1: Forgetting closing tags

Incorrect:

```
<p>This paragraph is not closed.
```

Correct:

```
<p>This paragraph is closed.</p>
```

Mistake 2: Writing another opening tag instead of a closing tag

Incorrect:

```
<h1>My Page<h1>
```

Correct:

```
<h1>My Page</h1>
```

Mistake 3: Putting content inside void elements

Incorrect:

```
<img>Broken image</img>
```

Correct:

```

```

Mistake 4: Reusing the same `id`

Incorrect:

```
<h1 id="title">Cats</h1>  
<h1 id="title">Dogs</h1>
```

Correct:

```
<h1 id="cat-title">Cats</h1>  
<h1 id="dog-title">Dogs</h1>
```

Mistake 5: Using spaces inside `id` values

Incorrect:

```
<h1 id="main heading">My Page</h1>
```

Correct:

```
<h1 id="main-heading">My Page</h1>
```

Mistake 6: Misusing boolean attributes

Incorrect:

```
<input type="checkbox" checked="false">
```

Correct if checked:

```
<input type="checkbox" checked>
```

Correct if not checked:

```
<input type="checkbox">
```

Mistake 7: Repeating the same class on the same element

Technically valid but unnecessary:

```
<div class="card blue card"></div>
```

Better:

```
<div class="card blue"></div>
```

28. Debugging Sprint 1 Broken Code

Broken code:

```
<h1 id="main heading">My Page<h1>
<img>Broken image</img>
<div class="card blue card"></div>
<input type="checkbox" checked="false">
```

Now fix each problem.

Problem 1: Broken Heading

Broken:

```
<h1 id="main heading">My Page<h1>
```

Problems:

- The `id` has a space.
- The closing `h1` tag is wrong.

Fixed:

```
<h1 id="main-heading">My Page</h1>
```

Why this is correct:

- `main-heading` does not contain a space.
 - `</h1>` correctly closes the heading.
-

Problem 2: Broken Image

Broken:

```
<img>Broken image</img>
```

Problems:

- `img` is a void element.
- `img` cannot contain text.
- `img` should have `src`.
- `img` should have `alt`.

Fixed:

```

```

Why this is correct:

- It has no closing tag.
- It has a `src` attribute.
- It has an `alt` attribute.

Problem 3: Repeated Class

Original:

```
<div class="card blue card"></div>
```

This is not the worst error, but it is unnecessary.

The class `card` appears twice.

Better:

```
<div class="card blue"></div>
```

Why this is better:

- The repeated class was removed.
- The element still has two useful classes: `card` and `blue`.

Problem 4: Misleading Boolean Attribute

Broken:

```
<input type="checkbox" checked="false">
```

Problem:

- `checked` is a boolean attribute.
- If `checked` is present, the checkbox is checked.

- The value `false` does not turn it off.

Fixed if you want it checked:

```
<input type="checkbox" checked>
```

Fixed if you do not want it checked:

```
<input type="checkbox">
```

Final Fixed Version

```
<h1 id="main-heading">My Page</h1>



<div class="card blue"></div>

<input type="checkbox">
```

This fixed version follows Sprint 1 rules:

- The heading has a correct closing tag.
- The `id` has no spaces.
- The image is written as a void element.
- The image has `src` and `alt`.
- The repeated class is removed.
- The checkbox does not misuse `checked="false"`.

29. Small Beginner Project

Create a file named:

```
index.html
```

Then write this HTML:


```
<h1 id="page-title">My Favorite Animal</h1>

<div class="animal-card favorite">
  <h2>Cats</h2>

  <p>Cats are small animals that many people keep as pets.</p>

  <p>I like cats because they are quiet, curious, and independent.</p>

  

  <p>
    <input type="checkbox" checked>
    I have read about this animal.
  </p>
</div>
```

30. Explanation of the Beginner Project

Heading

```
<h1 id="page-title">My Favorite Animal</h1>
```

This creates the main heading.

It also gives the heading a unique `id`:

```
page-title
```

Container

```
<div class="animal-card favorite">
```

This creates a generic container.

It has two classes:

```
animal-card  
favorite
```

The closing tag is:

```
</div>
```

Section Heading

```
<h2>Cats</h2>
```

This creates a smaller heading inside the animal card.

Paragraphs

```
<p>Cats are small animals that many people keep as pets.</p>  
  
<p>I like cats because they are quiet, curious, and independent.</p>
```

These create two paragraphs.

Each paragraph has:

- Opening `<p>` tag
- Text content
- Closing `</p>` tag

Image

```

```

This displays an image.

It has:

```
src="cat.jpg"
alt="A gray cat sitting on a wooden chair."
```

It does not have a closing tag because `img` is void.

Checkbox

```
<input type="checkbox" checked>
```

This creates a checkbox.

The `checked` attribute means the checkbox starts checked.

The `input` element is void, so it does not have a closing tag.

31. Sprint 1 Practice Task

Build this from memory:

```
<h1 id="profile-title">My Profile</h1>

<div class="profile-card beginner">
  <h2>About Me</h2>

  <p>My name is Sangeeth.</p>

  <p>I am learning HTML fundamentals.</p>

  

  <p>
    <input type="checkbox" checked>
    I completed Sprint 1.
  </p>
</div>
```

32. Practice Task Checklist

After writing the practice task, check these points:

No duplicate id.
No spaces inside id.
Every paragraph has `</p>`.
Every heading has a closing tag.
The `img` has `src` and `alt`.
The `img` has no closing tag.
The checkbox uses `checked` intentionally.
Every non-void element is properly closed.
Void elements do not contain content.

33. Sprint 1 Revision Questions

Use these questions to test yourself.

Question 1

What does HTML control on a web page?

Answer:

HTML controls the structure and content of a web page.

Question 2

What is an opening tag?

Answer:

An opening tag starts an element.

Example:

`<p>`

Question 3

What is a closing tag?

Answer:

A closing tag ends an element.

Example:

```
</p>
```

Question 4

Why does `img` not have a closing tag?

Answer:

Because `img` is a void element. Void elements cannot contain content and do not need closing tags.

Question 5

Where do attributes go?

Answer:

Attributes go inside the opening tag.

Example:

```
<p class="intro">Hello</p>
```

Question 6

What does `src` do on an image?

Answer:

The `src` attribute tells the browser where the image file is.

Question 7

What does `alt` do on an image?

Answer:

The `alt` attribute gives a text description of the image.

Question 8

Why is `checked="false"` still a problem?

Answer:

Because `checked` is a boolean attribute. If it is present, the checkbox is checked, even if the value says `false`.

Question 9

When should you use `id`?

Answer:

Use `id` for one unique element on the page.

Question 10

When should you use `class`?

Answer:

Use `class` for reusable grouping. Many elements can share the same class.

34. Sprint 1 Key Rules

Memorize these rules:

HTML gives structure and content.
Most elements have an opening tag and closing tag.
Closing tags need a slash.
Elements should be nested correctly.
Use lowercase tag names.
Void elements do not have closing tags.
Void elements do not contain content.
Attributes go inside opening tags.
Most attributes use name="value" syntax.
img needs src and alt.
Boolean attributes are on when present.
id should be unique.
class can be reused.
id values should not contain spaces.

35. Sprint 1 Final Self-Check

You are ready to move on from Sprint 1 when you can do all of these without looking at notes:

- Explain what HTML does.
- Write a heading element.
- Write a paragraph element.
- Correctly close non-void elements.
- Correctly nest one element inside another.
- Write lowercase tag names.
- Explain what a void element is.
- Write an `img` element with `src` and `alt`.
- Write an attribute using `name="value"` syntax.
- Explain why `checked="false"` is wrong.
- Use `checked` correctly.
- Use `id` for one unique element.
- Use `class` for reusable groups.
- Find and fix simple broken HTML.

36. Mini Glossary

HTML

The language used to structure content on a web page.

Element

A complete piece of HTML, usually made of an opening tag, content, and closing tag.

Example:

```
<p>Hello</p>
```

Opening Tag

The tag that starts an element.

Example:

```
<p>
```

Closing Tag

The tag that ends an element.

Example:

```
</p>
```

Content

The text or child elements inside an element.

Example:

```
<p>This is content.</p>
```

Nesting

Putting one element inside another element.

Example:


```
<div>
  <p>Nested paragraph.</p>
</div>
```

Void Element

An element that does not contain content and does not have a closing tag.

Example:

```

```

Attribute

Extra information added to an element inside the opening tag.

Example:

```
<p class="intro">Hello</p>
```

Boolean Attribute

An attribute that is turned on by being present.

Example:

```
<input type="checkbox" checked>
```

id

A unique label for one element.

Example:

```
<h1 id="page-title">Page Title</h1>
```

class

A reusable label for one or more elements.

Example:

```
<div class="card">Card content</div>
```

src

An attribute that tells the browser where an image file is.

Example:

```

```

alt

An attribute that describes an image in text.

Example:

```

```

37. Final Sprint 1 Example

This example combines the main Sprint 1 concepts:

```
<h1 id="animal-page-title">My Favorite Animals</h1>

<div class="animal-card cat-card">
  <h2>Cats</h2>

  <p>Cats are quiet, curious animals.</p>

  

  <p>
    <input type="checkbox" checked>
  </p>
</div>
```

```

    I like cats.
  </p>
</div>

<div class="animal-card dog-card">
  <h2>Dogs</h2>

  <p>Dogs are friendly and energetic animals.</p>

  

  <p>
    <input type="checkbox">
    I have read about dogs.
  </p>
</div>

```

This example includes:

- One unique `id`
- Reusable classes
- Multiple classes on elements
- Proper headings
- Proper paragraphs
- Correct nesting
- Images with `src` and `alt`
- Void elements written correctly
- Boolean attribute `checked` used correctly
- No duplicate `id`
- No spaces inside `id`

38. Sprint 1 Completion Standard

Sprint 1 is complete when you can write simple HTML correctly from memory.

You should be able to look at this broken code:

```

<h1 id="main heading">My Page<h1>
<img>Broken image</img>
<input type="checkbox" checked="false">

```

And fix it to:

```
<h1 id="main-heading">My Page</h1>  
  
<input type="checkbox">
```

You should also be able to explain why each fix is correct.

That is the main point of Sprint 1.